

TOMÁŠ HUTŇAN

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Slovak National · Brno, Czech Republic

SELECTED EDUCATION

MASARYK UNIVERSITY | *BSc Informatics* Sep 2024 – Present

- Academic standing: 95th percentile of cohort.
- Relevant coursework: Machine Learning, Statistics, Formal Languages, Modelling & Simulations, Image Processing.

EXPERIENCE

MASARYK UNIVERSITY | *Faculty of Informatics* Brno, Czechia

Research Assistant Oct 2026 – Present

- Research LLM safety and clinical guardrails within Project EMPOWER.
- Generate synthetic oncology datasets to stress-test model boundaries.

Teaching Assistant Sep 2025 – Present

- Instruct weekly seminars in Programming, Algorithms, and Intro to AI for 20+ students.
- Review 100+ submissions weekly, coaching students on maintainable software practices.

CHYRONHEGO CZECH S.R.O.

Software QA & Automation Intern Nov 2024 – Dec 2025

- Engineered Python automated testing infrastructure for high-availability systems.
- Authored and integrated 1,000+ test cases, replacing manual regression workflows.
- Reviewed 100+ pull requests and proposed code improvements adopted into production.

SELECTED PROJECTS

CUDA IMAGE ENHANCEMENT | *GPU Engineer — C++ / CUDA* Mar – May 2026

- Developed a GPU-accelerated electron-microscopy image pipeline (25+ FPS) with TESCAN.
- Optimized CUDA kernels (shared memory, coalescing) for denoising and deconvolution.

TATRASHARE | *Technical Advisor — React Native / MongoDB* Apr 2026

Winner: MLH Best Use of MongoDB & Tatra Banka Challenge

- Architected an AI-powered finance platform for real-time expense splitting.
- Integrated Auth0 authentication and scalable backend data structures (MongoDB Atlas).

TEARZ | *Project Lead — ML / Computer Vision* Apr 2026

Honorable Mention: Hack Košice 2026

- Led team to build a biomedical ML classification pipeline, achieving a 93% F1 score.
- Engineered preprocessing for noisy tear microscopy images, extracting 300+ features.

1st Place Overall: Newsmatics Hackathon 2025

- Built a financial market analytics platform processing unstructured text and market data.
- Designed NLP pipelines and implemented transformer models for consensus analysis.

SKILLS

Core Stack Python, C++, PyTorch, Docker, Linux, OpenCV, Pandas

Also Used TypeScript, C#, FastAPI, PostgreSQL, MongoDB, React

Specializations Computer Vision, NLP, GPU Optimization, Automated Testing

Languages English (C1/C2), German (A1/A2), Slovak (Native)